

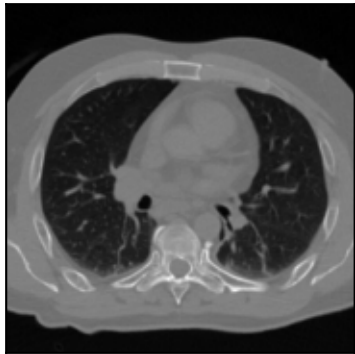
DVF QA pipeline report

DIR-Lab-case4

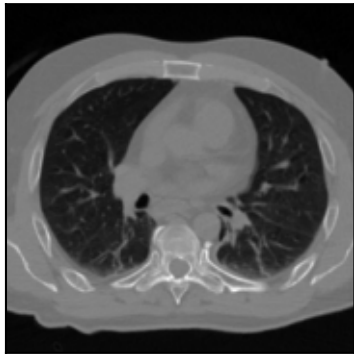
Field	Value
Case	DIR-Lab-case4
Generated	2026-05-20 09:24:51
source	Case4Pack
fps	15.0
n_cycles	4
frames_per_cycle	30
n_phases	10
ct_shape	(99, 256, 256)
spacing_xyz_mm	(1.13, 1.13, 2.5)
detector_shape	(256, 256)
pixel_spacing_mm	(1.0, 1.0)
device	cpu

4DCT phase preview

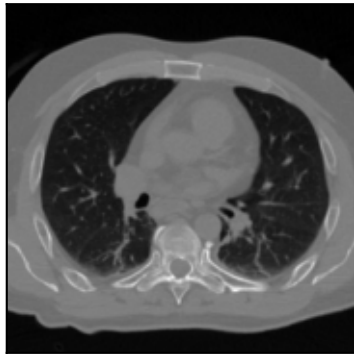
ϕ_0



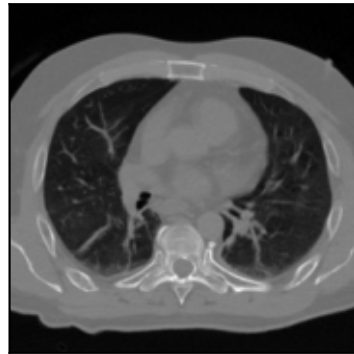
ϕ_1



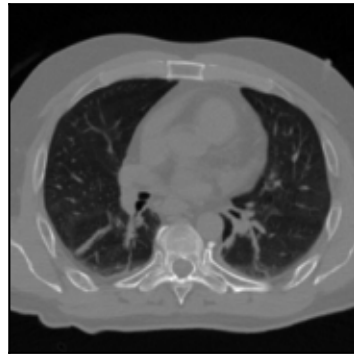
ϕ_2



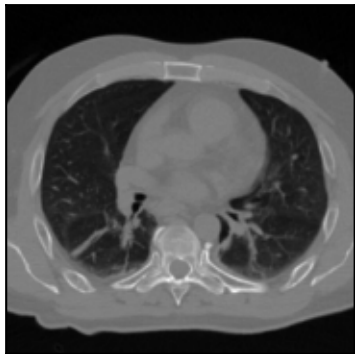
ϕ_3



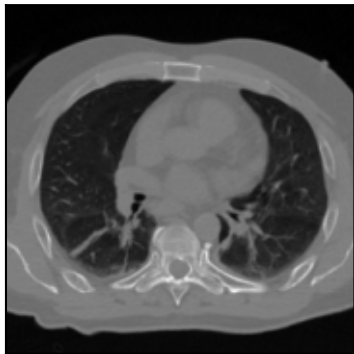
ϕ_4



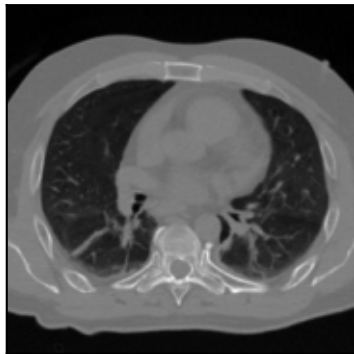
ϕ_5



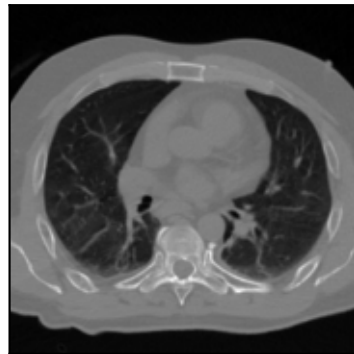
ϕ_6



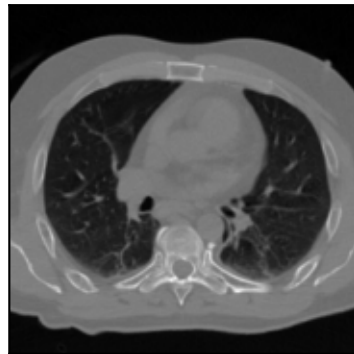
ϕ_7



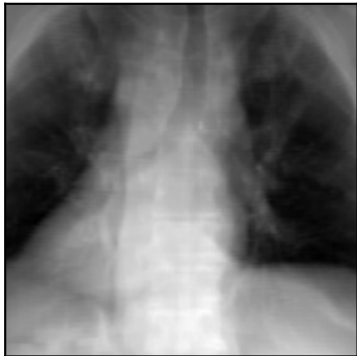
ϕ_8



ϕ_9



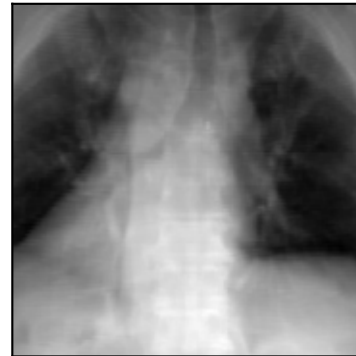
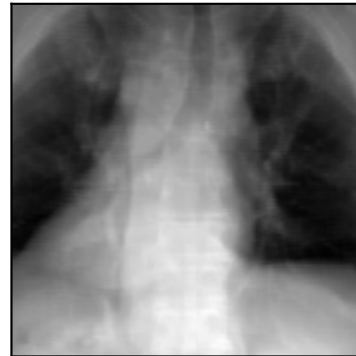
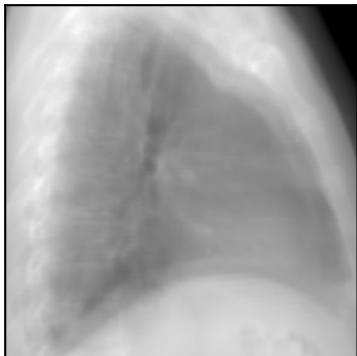
Per-phase DRRs

AP φ_0 AP $\phi 1$ AP ϕ_2 AP $\phi 3$ 

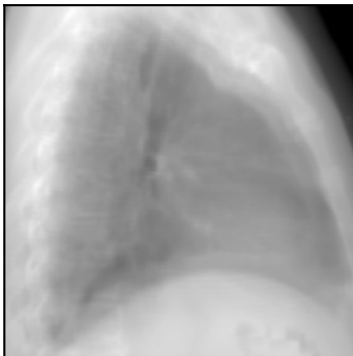
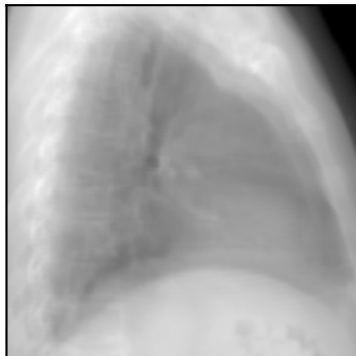
AP ϕ4

AP $\phi 5$ AP ϕ_6 

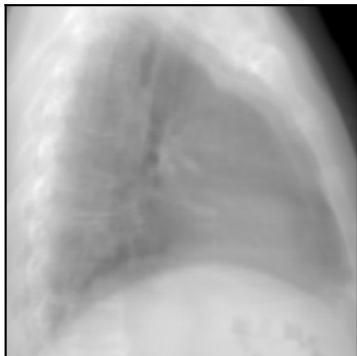
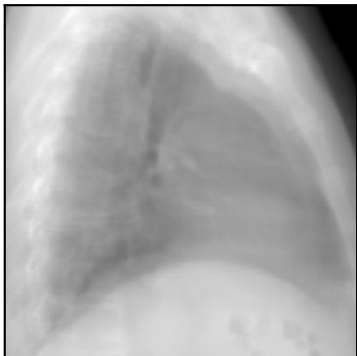
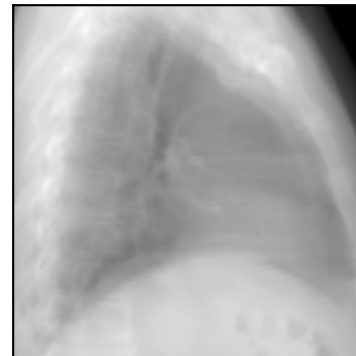
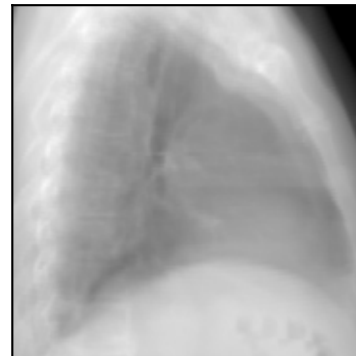
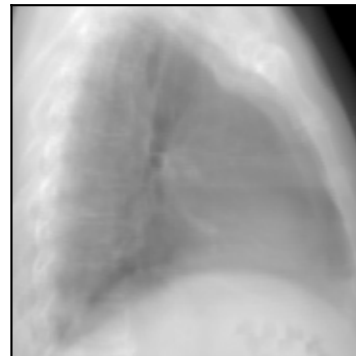
AP φ7

AP $\phi 8$ AP φ_9 Lat φ_0 

Lat ϕ_1

Lat ϕ_2 

Lat ϕ_3

Lat ϕ_4 Lat ϕ_5 Lat ϕ_6 Lat ϕ_7 Lat ϕ_8 Lat ϕ_9 

Dynamic sequence samples

AP t=0 (0.00s)



AP t=24 (1.60s)



AP t=48 (3.20s)



AP t=72 (4.80s)



AP t=96 (6.40s)



AP t=120 (8.00s)



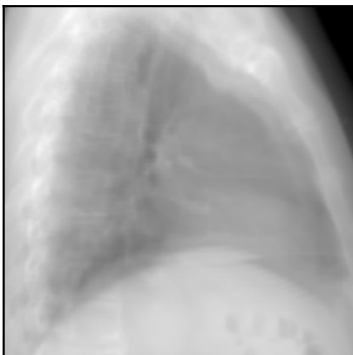
Lat t=0



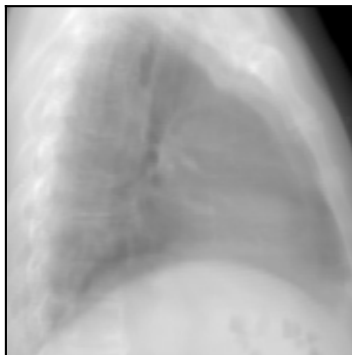
Lat t=24



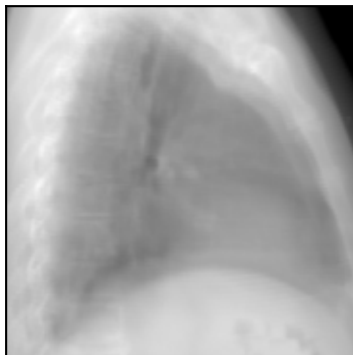
Lat t=48



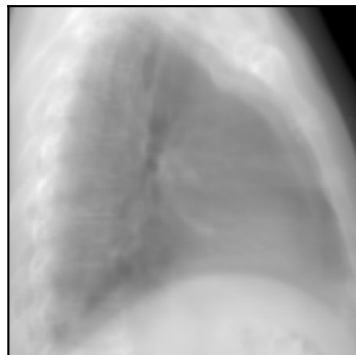
Lat t=72



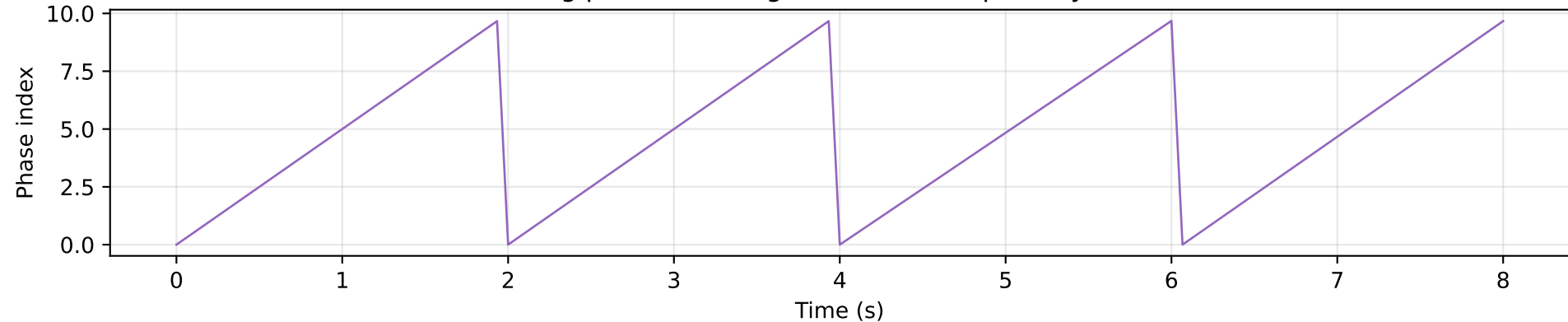
Lat t=96



Lat t=120

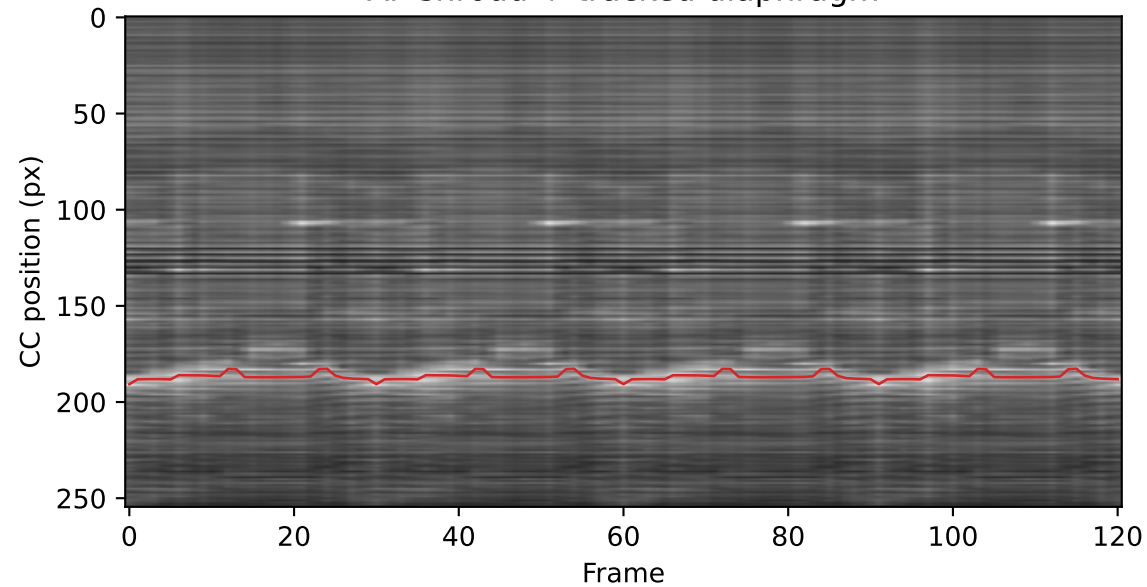


Driving phase curve
Driving phase curve (ground-truth respiratory motion)

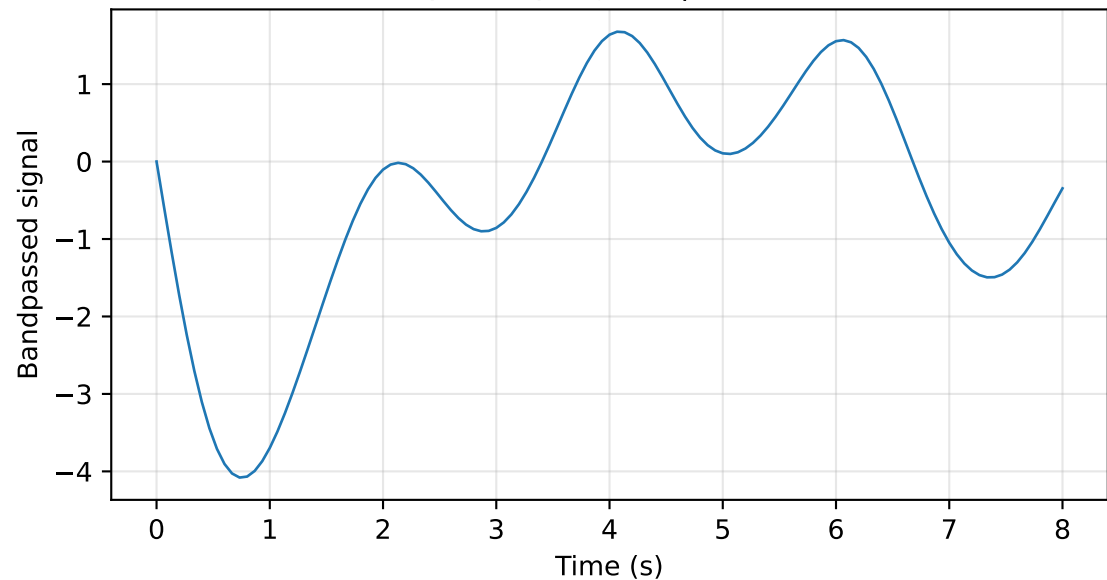


Amsterdam Shroud — AP

AP shroud + tracked diaphragm

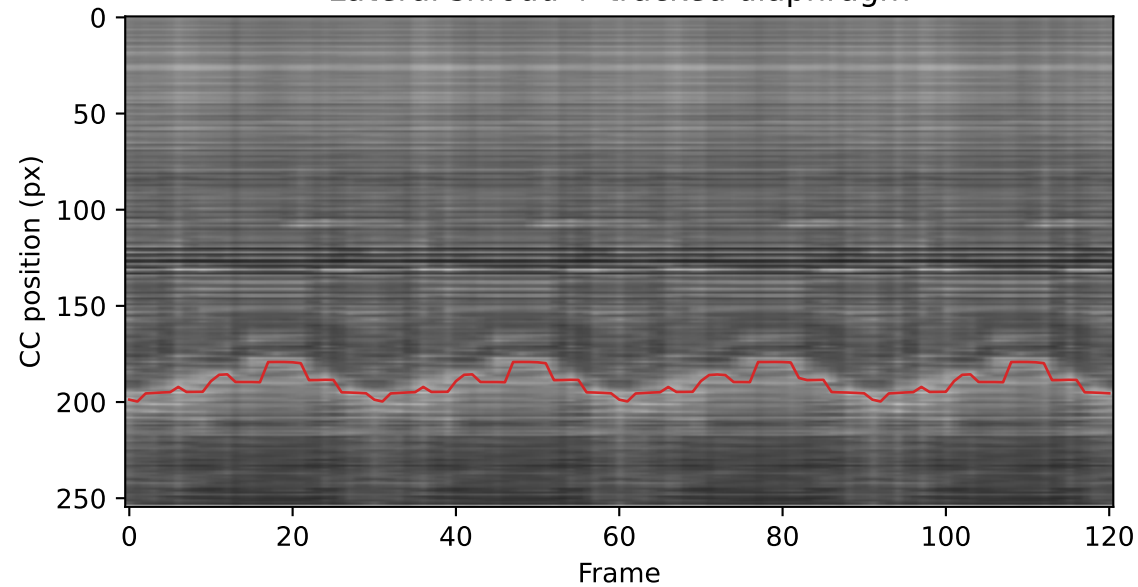


AP respiratory signal | $f=0.124$ Hz

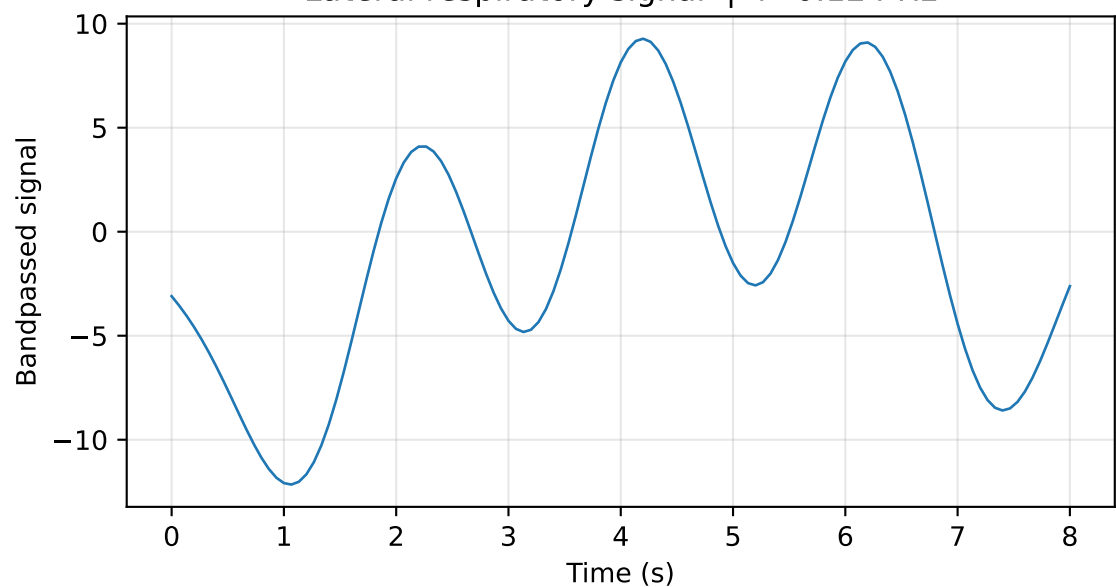


Amsterdam Shroud — Lateral

Lateral shroud + tracked diaphragm

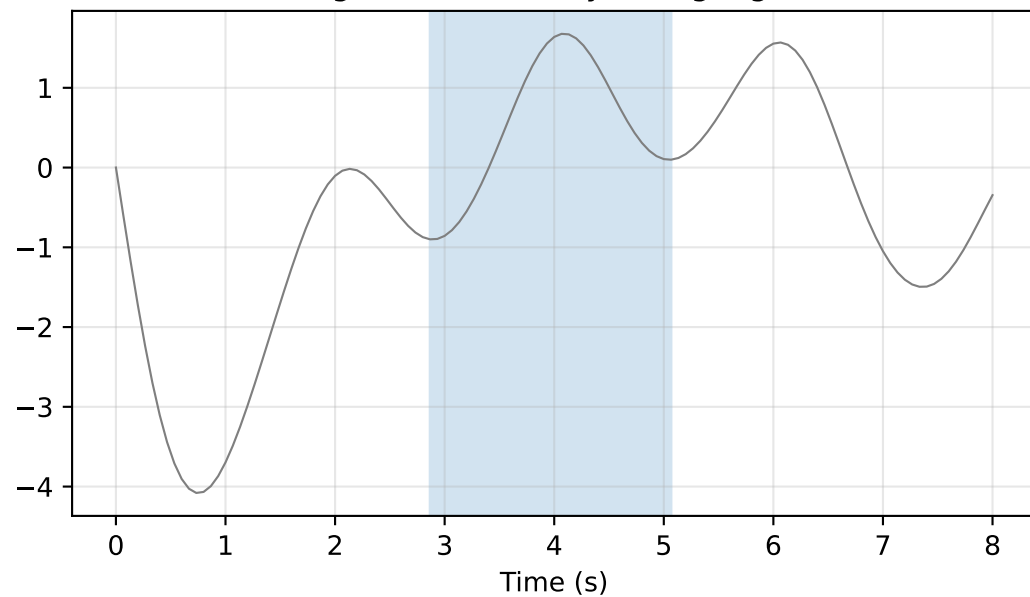


Lateral respiratory signal | $f=0.124$ Hz

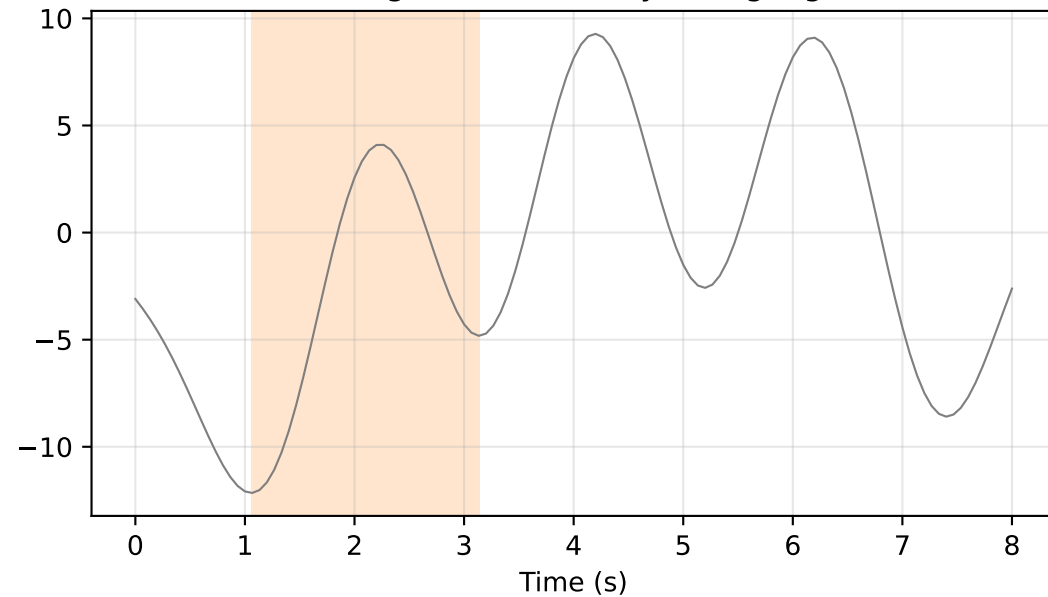


Cycle pairing

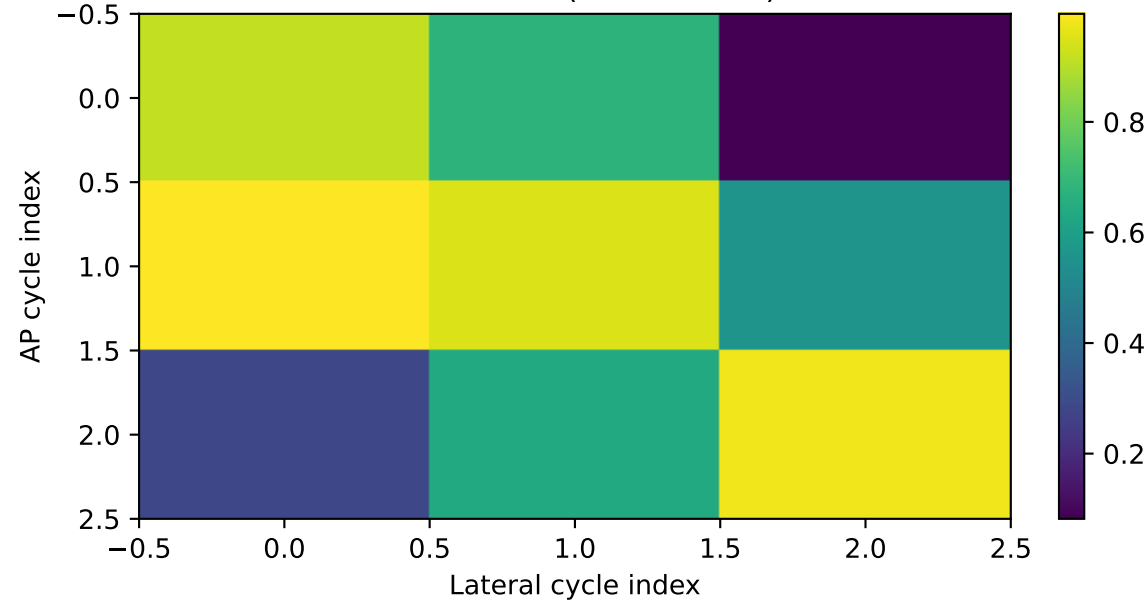
AP signal (selected cycle highlighted)



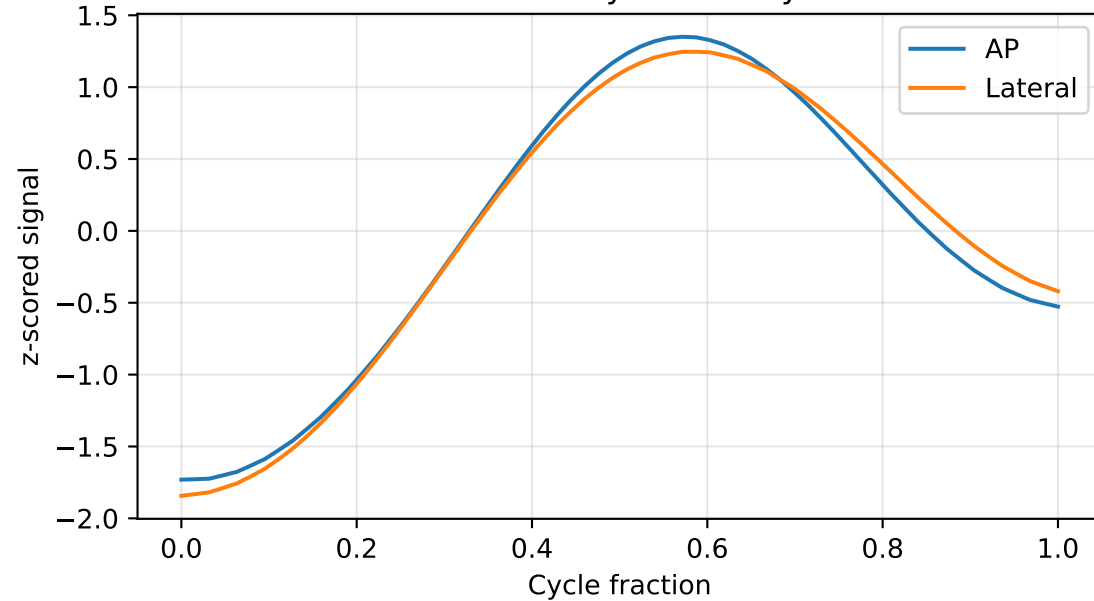
Lateral signal (selected cycle highlighted)



NCC matrix (best=0.995)



Selected cycle overlay



Metrics

Metric	Value
AP dominant frequency (Hz)	0.1240
AP period (frames)	121.00
Lateral dominant frequency (Hz)	0.1240
Lateral period (frames)	121.00
Best pair NCC	0.9955
AP cycle frames	43 - 76
Lateral cycle frames	16 - 47
min_period_s	1.0000